PRETTY TOXIC

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DESIGNING WITH A TOXIC INVASIVE SPECIES
TOWARDS ECOLOGICAL REGENERATION

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DEGREE PROJECT
MASTER OF ARTS IN DESIGN ECOLOGIES, 120 CREDITS

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STOCKHOLM 2026

ABSTRACT

Invasive alien species are one of the leading causes of global biodiversity loss and a critical ecological and economical challenge.
The toxic Giant hogweed is one of the most aggressively spreading *Invasive alien species of Union concern*, threatening native ecosystems and human health. As a result of ongoing management efforts, large amounts of this fast-growing plants are cut down annually. The biomass is discarded, despite their high cellulose content.

The project Pretty Toxic responds to this paradox by connecting design with environmental protection, exploring the potential of Giant hogweed as a alternative resource for pulp production and material applications. Grounded in the principles of material-driven design (MDD), the research focuses on detoxifying and processing stems of Giant hogweed into non-wood pulp, which is subsequently formed into three-dimensional shapes through a wet-molding process. Within this framework, the design concept is based on the absorbent properties of hogweed-derived cellulose in combination with wool, examining how these materials can be systematically integrated into spatial contexts as responsive architectural elements. The resulting collection, Pretty Toxic, introduces geometric tiles designed to transform walls and ceilings into active surfaces capable of absorbing sound, moisture, particulate matter and gaseous air pollutants. Arranged in composition, the tiling creates variable monohedral patterns that enhance the spatial performance while referencing the invasive growth of the toxic species.

This thesis reflects on present and future possibilities of underutilized materials and examines implications of positioning Giant hogweed as a material resource. In doing so, it demonstrates how invasive alien species may be reframed as materials with ecological and economic value, contributing to a regenerative material system that reduces the environmental impacts associated with the conventional cellulose industry.

INVASIVE ALIEN SPECIES, GIANT HOGWEED, MATERIAL-DRIVEN DESIGN (MDD), DESIGN RESEARCH, ALTERNATIVE CELLULOSE RESOURCES, MONOHEDRAL GEOMETRIES, SOUND ABSORPTION, AIR PURIFICATION

ACKNOWLEDGEMENTS

I like to express my sincere gratitude to everyone who supported, guided, and contributed to this thesis throughout the process.

First, i like to thank Konstfack University of Arts, Crafts and Design and my thesis advisor, Jenny Althoff, for her invaluable guidance, encouragement, and thoughtful feedback throughout the project. I am also grateful to my tutors, Carl-Johan, Martin Avila and Cecilia Wahlberg, for their support, insights, and continuous engagement during the development of this work.

My thank goes to Harald Cederlund at the Swedish University of Agricultural Sciences and Fredrik Fogelberg at RISE Research Institutes of Sweden for sharing their research expertise and valuable knowledge, which greatly contributed to this study. My gratitude also goes to Anders Kronham at SRV for providing practical insights and support connected to the project, and to Mickael Lindström for his valuable discussions and guidance within material research.

A sincere thank you goes to Gunnar Söder for providing the freedom and trust to explore the material process, as well as for his support in the workshops, access to equipment, and guidance throughout the experimentation and making process.

Gratitude is also extended to my classmates for their encouragement, feedback, and inspiring discussions throughout this journey. A special thank you goes to Francesca Pezotti and Soraya Hassan for their continuous support throughout this process.

Lastly, I would like to acknowledge my friends and family for their encouragement, patience, and support throughout this project.

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DESIGNING FOR BIODIVERSITY

Materials are deeply connected with our lifestyle, culture and design. Historically, many communities developed material cultures rooted in local ecologies, working with what was available to them and fostering regenerative relationships with their environments. Today, design often relies on globally standardized materials. Yet those materials are sourced through extractive global supply chains and industrial production that intensify ecological degradation, biodiversity loss, and climate pressures. The apparent abundance conceals their dependence on finite resources, complex logistics, and energy-intensive processing. But, in a future increasingly shaped by global warming and biodiversity loss, what happens if these materials are no longer available?

This project began with the aim to disengage from conventional materials systems and challenging existing perceptions of value that shape current design practices. *But, how to design for biodiversity?* Instead of simply reducing environmental impact, I aimed to propose a holistic approach to ecosystem regeneration and resilience, starting with the identification of alternative, resilient material resources. During this search, I encountered invasive alien species: organisms often perceived solely as ecological threats, yet characterized by resilience and adaptability under changing environmental conditions. This tension became the foundation of the project.

The Giant hogweed is a toxic plant recognized as one of Europe's most aggressive invasive species. This project explores the potential of this species as overlooked material resource, shifting it from an ecological threat towards a valuable material opportunity. In doing so, Pretty Toxic aims to reframe the current narrative of the species while envisioning a design practice that actively contributes to the restoration and protection of ecosystems. This raises the central research question:

HOW TO DESIGN WITH A TOXIC INVASIVE ALIEN SPECIES AND CAN IT BE PART OF OUR CONTEMPORARY MATERIAL ECONOMY?

PROJECT POSITIONING

This degree project is positioned within design, exploratory material research and the research field of alternative cellulose resources, which investigates plant fibers as substitutes for conventional industrial resources.

Following a Material Driven Design approach, the research explores how Giant hogweed biomass, typically removed or discarded as waste, can be sourced, processed, and transformed into a material application without contributing to further ecological imbalance. Rather than introducing new cultivated resources, Pretty Toxic investigates how existing ecological disturbances can be reframed as material opportunities.

Through a systemic approach, the project situates invasive biomass within broader relationships between ecological management, material development, and design application. It explores how invasive biomass can be harvested, processed, and integrated into material systems while supporting biodiversity management and ecosystem protection.

Pretty Toxic contributes to ongoing research on alternative cellulose resources by expanding the scope of non-wood cellulose sources from agricultural crops to invasive plant species and biomass generated through ecological management. Furthermore, it contributes to the emerging exploration of Giant hogweed within scientific and design research by investigating the plant beyond its ecological impact, focusing on its potential as functional material. Within this context, the project proposes a tile collection that translates the developed material into a specific context of use within interior environments, demonstrating the value of overlooked materials and addressing human needs through design.

INVASIVE ALIEN SPECIES

The shifts of the climate and ecosystems connected to it, is not a distant scenario anymore. It is actively changing life on Earth, which we already can measure, observe and feel.

Plants migrating and shifting their geographic ranges has always been part of natural ecological processes occurring over generations. However, the rate and scale of species movement, distribution and biodiversity loss has been increased, due to environmental changes driven by climate change and the impact of human activities. The five primary drivers are habitat loss and degradation, overexploitation of species, invasive alien species, climate change, and pollution of air, water, and soil with chemicals, plastics, and other contaminants[1].

As native species decline or disappear across different bioregions, other species increasingly establish themselves, some of which are classified as Invasive Alien Species (IAS). Alien species are plants or animals introduced by humans into regions outside their native range. Some of these species are becoming “invasive”, because of their fast expansion and negative impact on ecosystems.[2] The fast distribution of invasive species can change ecosystem functions, reduce ecological resilience, and limit the availability of natural resources for human and non-human. According to World Wide Fund for Nature (WWF), Invasive Alien Species (IAS) represent the second greatest threat to global biodiversity.[3]

[1] World Wide Fund for Nature (WWF), Living Planet Report 2024: A System in Peril (Gland, Switzerland: WWF, 2024), https://wwflpr.awsassets.panda.org/downloads/2024-living-planet-report-a-system-in-peril.pdf.

[2] International Union for Conservation of Nature (IUCN), Invasive Alien Species and Climate Change (February 2025 update), accessed February 7, 2026, https://iucn.org/sites/default/files/2025-03/invasive-alien-species-and-climate-change-feb.-2025-update.pdf.

[3] World Wide Fund for Nature (WWF), “12 New Invasive Alien Species Included to Europe’s List,” accessed February 7, 2026, https://www.wwf.eu/?303370/12-new-invasive-alien-species-included-to-Europes-list.



FIG. 1
Dried Giant Hogweed
Source: Author



FIG. 2
Flowering Giant hogweed
Source: https://environmentcontrols.co.uk/what-does-giant-hogweed-look-like/





FIG. 3
Documentation of Giant hogweed in Hagaparken, Stockholm
Source: Author

THE GIANT HOGWEED

The Giant hogweed (Heracleum mantegazzianum) is a toxic flowering plant native to the mountain landscape of the Caucasus mountains. It is characterized by its height up to five meters, large leaves and an umbrella-shaped inflorescence. In the 19th century it was introduced to Europe by botanists and landscape designers as an exotic ornamental curiosity, decorating European gardens. However, due to its high seed production, rapid growth, and resilience, the species began to spread beyond cultivation. Since then, it established itself aggressively in urban areas and along riverbanks, posing an risk for native vegetation.

Its height and large leaves allow Giant hogweed to overshadow surrounding plants, blocking sunlight and suppressing the growth of native species. It forms dense stands that create monocultures, significantly reducing plant diversity and limiting habitat availability for insects and other organisms across large areas. Additionally, the plants develop an extensive root system, which can contribute to soil erosion when the plant dies back after flowering.[4]

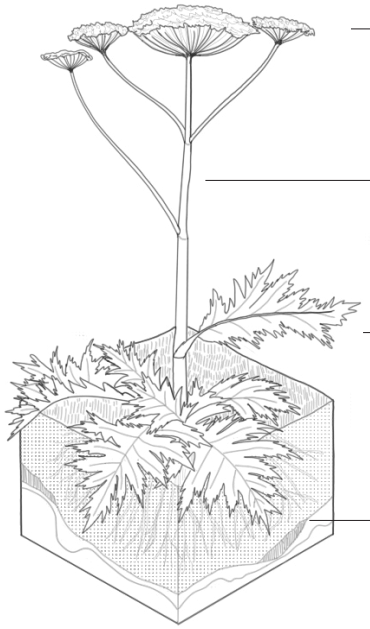
Throughout it's lifecycle, hogweed produces a photo-toxic sap containing furanocoumarins, which function as a powerful natural defense mechanism and allows the species to rapidly dominate new areas. The toxins are activated by sunlight and cause severe skin burns on contact, making the plant a threat to human health and well-being.[5]

As a result, since 2017, Giant hogweed is listed as *Invasive Alien Species of Union Concern* under the European Union's Regulation 1143/2014. This designation legally obliges countries within the EU to implement control strategies to manage the distribution of the species. Responsible authorities in Sweden includes the Swedish Environmental Protection Agency, which classified Giant hogweed as a high-risk species.[6]

[4] European and Mediterranean Plant Protection Organization (EPPO) (2009). EPPO data sheet on Invasive Alien Plants: Heracleum mantegazzianum, Heracleum sosnowskyi and Heracleum persicum. EPPO Bulletin 39, 489–499.

[5] Charlotte Nielsen, Hans Peter Ravn, Wolfgang Nentwig, and Max Wade, The Giant Hogweed Best Practice Manual: Guidelines for the Management and Control of an Invasive Weed in Europe, Forest & Landscape Denmark 44 (Hoersholm, Denmark: Forest & Landscape Denmark, 2005), 22.

[6] SLU Artdatabanken, "Jätteloka (Heracleum mantegazzianum)," Artfakta, accessed March 16, 2026, https://artfakta.se/taxa/219680.



FAST DISTRIBUTION

high seed production

FAST GROWTH

growing up to 5 m in one year

OVERGOWING NATIVE SPECIES

leaves create a thick canopy that blocks sunlight

SOIL DEGRADATION

deminishes ground cover and altering soil composition, causing soil erosions and habitat loss

FIG. 4
Morphology and properties of the Giant hogweed
Source: Author

CONTROLLING HOGWEED

Recommended management strategies depend on the size, density, and location of the population. For small populations, mechanical methods such as root cutting and removal are often effective, while larger infestations may require repeated mowing, chemical treatments, or biological control methods, including grazing by herbivores. Some animals, such as goats, sheep, and cattle, show higher tolerance to the plant's phototoxic compounds and can be used as a grazing strategy, reducing the risk of direct exposure for workers.[7] According to the Swedish Environmental Protection Agency, mechanical control methods should be prioritized when managing invasive alien plants. Chemical plant protection products should be avoided as far as possible due to their potential impacts on nature and the environment.[8]

For this project, the control strategies are essential to mitigate risks throughout the exploration phase and to anticipate. An interview with researchers Harald Cederlund (SLU) and Fredrik Fogelberg (RISE), who published their study on control methods of Giant hogweed in Sweden in 2025, highlighted that in the Stockholm area, management is decentralized and falls under the responsibility of municipalities and private landowners.[9]
A following field trip to SRV Återvinning AB in Huddinge provided insight into the practical challenges of managing Giant hogweed.

Based on this analysis, the key challenges are that the removal of dispersed Giant hogweed populations is labour- and cost-intensive, requiring careful handling and repeated interventions. In addition, occurrences are often widely scattered, with limited updated mapping and systematic surveillance. As a result, Giant hogweed remains primarily framed through its negative impacts: as an ecological threat, an economic burden, and a persistent management challenge.

[7] Nielsen et al., The Giant Hogweed Best Practice Manual, 30-37.

[8] Naturvårdsverket, "Bekämpning av invasiva främmande växter på land," accessed February 15, 2026, https://www.naturvardsverket.se/vagledning-och-stod/invasiva-frammande-arter/bekampning-av-invasiva-frammande-vaxter-pa-land/bekampning-och-hantering-av-jatteloka.

[9] Naturvårdsverket, "Ansvarsfördelning för arbetet med invasiva främmande arter: Fastighetsägare och de som har nyttjanderätt till fastighet," accessed July 30, 2026, https://www.naturvardsverket.se/vagledning-och-stod/invasiva-frammande-arter/ansvarsfordelning-for-arbetet-med-invasiva-frammande-arter/fastighetsagare-och-de-som-har-nyttjanderatt-till-fastighet/.



FIG. 5
Interview and managing hogweed at SRV Huddinge
Source: Author

SHIFTING DISTRIBUTION

Despite ongoing management efforts to control Giant hogweed populations, the future distribution of the species remains uncertain as climate conditions continue to change. Giant hogweed is currently spreading across Central and Northern Europe, in regions with regular freezing periods. This is because, its reproduction is dependent on a cold period for its seeds to be activated. As global temperatures rise and environmental conditions shift, the plants future becomes uncertain. In the context of this research on Giant hogweed, this raises the question if the plants continue to grow.

Researchers have developed different climate change scenarios to understand the potential impacts of global warming on plant distributions. Based on the high-emission scenario RCP 8.5, with a projected temperature rise of over 4°C, ecosystems fundamentally change. This is why, milder winters are expected to alter Giant hogweed's ecological territory leading to a population decline of up to 94% across Central Europe by 2100. Simultaneously, SMHI predicts warmer and drier conditions in northern Europe, which extend growing seasons and a rise of groundwater levels in the mountains and inland of northern Sweden during winter. Since Giant hogweed thrives in moist, nutrient-rich soils, these climatic changes suggest that northern habitats are expected to remain suitable for resilient growth. In fact, the species is expected to expand by around 20% throughout northern territories.[10]

Such a transition of ecological territories illustrate how climate change is actively reshaping biodiversity and species distribution, not only of Giant hogweed but across ecosystems more broadly.

[10] Q. A. Anibaba, M. K. Dyderski, and A. M. Jagodziński, "Predicted Range Shifts of Invasive Giant Hogweed (Heracleum mantegazzianum) in Europe," Science of the Total Environment 825 (2022): 154053, https://doi.org/10.1016/j.scitotenv.2022.154053.

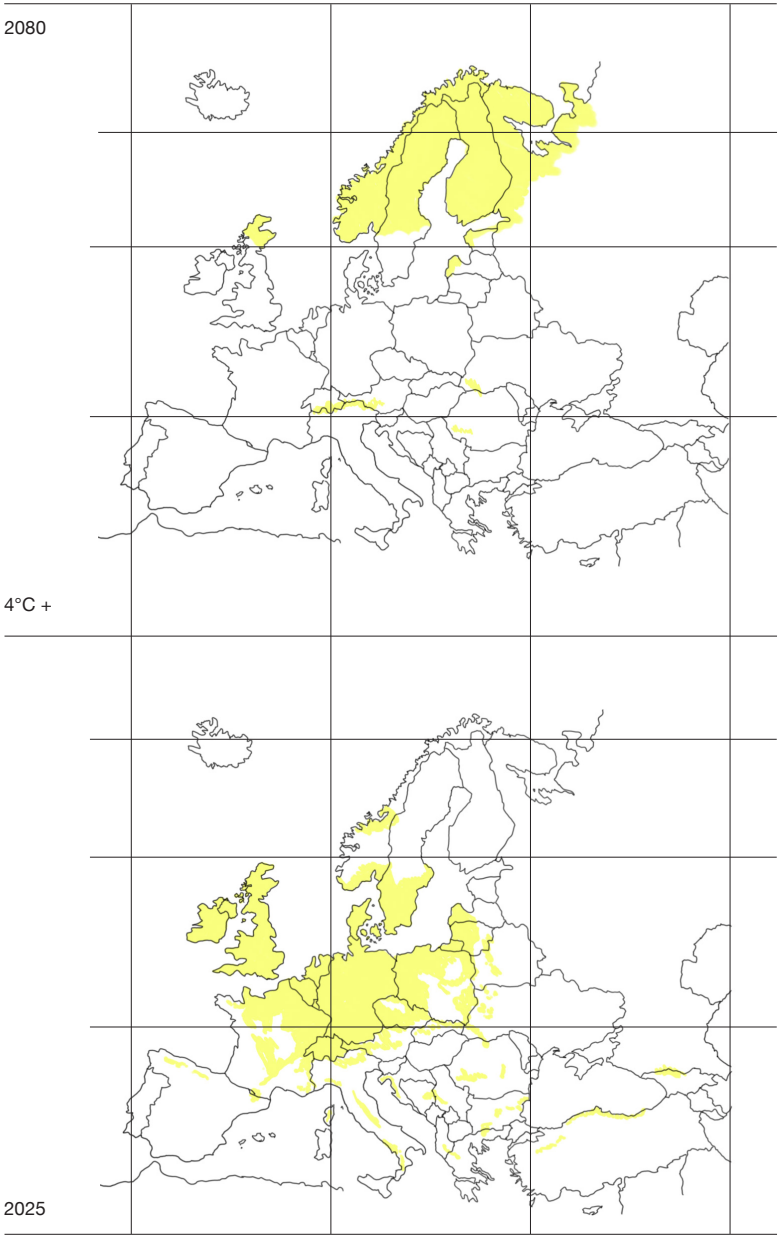


FIG. 6
Predicted range of distribution from 2025 to 2080
Source: Author

THE CELLULOSE INDUSTRY

The preceding chapters may raise the question:
Why should we design with a toxic species?

Cellulose is the main structural component of all plants and the foundation of a growing global industry. As a versatile bio-based material, it is used across sectors including paper, textiles, construction, pharmaceuticals, and cosmetics, with paper production accounting for roughly 46% of the cellulose market.[11]

The main source of industrial cellulose is wood, due to the high biomass availability of forests and the efficiency of established systems. However, the increasing global demand for cellulose places growing pressure on forest resources. Current extraction practices contribute to deforestation, intensive water and energy consumption, and greenhouse gas emissions [12], while climate change further threatens the resilience of forests. The boreal forest is already affected by 1.2°C, with further temperature increases causing irreversible damages and destruction.Ongoing global warming will exacerbate these effects, continuing to transform local ecosystems and with it the resources they provide. [13] *(FIG. 7)*

While wood has long been the dominant source of industrial cellulose, its not the only source providing cellulose-rich fibers. These challenges have accelerated research into alternative cellulose resources derived from agricultural residues, industrial by-products, and other non-wood biomass such as hemp, bagasse and flax.[14] A less explored category of alternative cellulose resources is invasive plant biomass. Similar to agricultural by-products, invasive biomass is typically removed and discarded despite containing valuable lignocellulosic fibres. The stems of Giant hogweed are composed of lignocellulose, similar to wood. This positions the species within the broader exploration of alternative cellulose sources and marking the starting point of this project Pretty Toxic.

[11] Market Reports World, "Cellulose Market Size, Share, Growth, and Industry Analysis, by Type (Commodity Cellulose Pulp, Cellulose Fibers, Cellulose Ethers, Cellulose), by Application (Paper, Textile & Apparel, Foods & Beverages, Pharmaceuticals, Cosmetics & Personal Care, Others), Regional Insights and Forecast to 2033," accessed March 29, 2026, https://www.marketreportsworld.com/market-reports/cellulose-market-14716468.

[12] Pulp and Paper Technology, "Fiber Other Than Wood: The Future of Sustainable Pulp and Paper," accessed July 30, 2026, https://www.pulpandpaper-technology.com/articles/fiber-other-than-wood-the-future-of-sustainable-pulp-and-paper.

[13] Dalia Kellou, Alexander Nauels, and Uta Klönne, Climate Impacts in Northern Forests (Climate Analytics, 2024), 2, https://ca1-clm.edcdn.com/publications/Climate-impacts-in-northern-forests.pdf?v=1714381265: 2.

[14] Pulp and Paper Technology, "Fiber Other Than Wood."

[15] Kellou, Nauels, and Klönne, Climate Impacts in Northern Forests, 24.

25 - 50 % OF BIODIVERSITY LOSS
OVER 3°C GLOBAL WARMING [15]

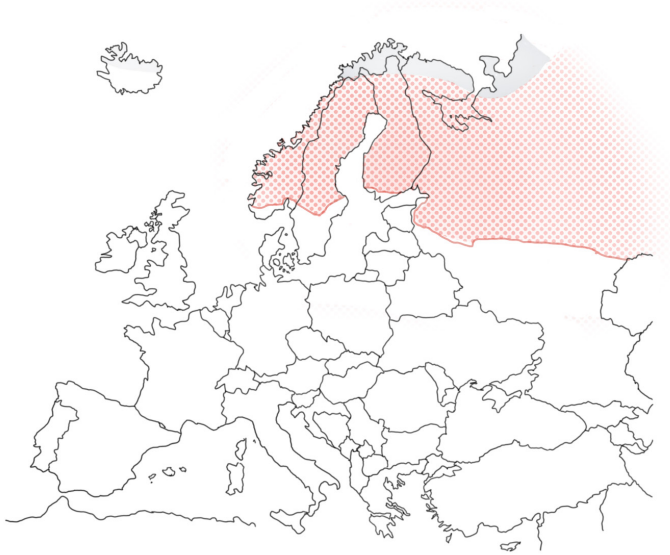


FIG. 7
Forest range and productivity
Source: Author

DESIGN REFERENCES

Despite the notoriety and distribution of Giant hogweed as an invasive species, research and accessible knowledge on the plants structure remain limited. Most existing studies focus on its ecological impacts, toxicity, and eradication, while its potential as a biomass resource has received comparatively little attention. This creates a gap between the extensive knowledge of the plant as an environmental problem and the limited exploration of its possibilities.

Within scientific research, interest in hogweed biomass is emerging. Related species, such as Sosnowsky's hogweed and Common hogweed are increasingly being investigated and used as biomass sources for fodder, bioethanol, cellulose, and biogas production.[16] Additionally, the Institute for Environmental Solutions (VRI) in Latvia has conducted research on Giant hogweed for biofuel production.[17] However, the application of the invasive species within design and material development remains largely unexplored. One exception is Atelier Schaft, a designer from the Netherlands, who has explored multiple dimensions of the plant. Throughout his practice, he created hogweed vases, seed boats and developed a material library, forming a series of experimental material applications. Ultimately, he presses the stems to create multi-coloured wall panels and conducts workshops exploring the material further.[18]

Viewing these references as a starting point, the project aims to expand this emerging research by exploring Giant hogweed as a non-wood cellulose resource and investigating how invasive biomass can be reinterpreted through material research and design.

[16] Liga Zvirbule-Berzina, Ilze Kļaviņa, and Māris Kļaviņš, "Invasive Species Application in Bioeconomy. Case Study Heracleum sosnowskyi Manden in Latvia," Energy Procedia 113 (2017): 238–239, https://doi.org/10.1016/j.egypro.2017.04.060.

[17] Vides risinājumu institūts, "Hogweed Mapping," accessed February 16, 2026, https://www.vri.lv/en/2010/04/hogweed-mapping/.

[18] Atelier Schaft, "Projects," accessed July 28, 2026, https://www.atelierschaft.nl.



FIG. 8
Predicted range of distribution from 2025 to 2080
Source: Author

HOW TO DESIGN WITH A TOXIC INVASIVE SPECIES?

RESEARCH THROUGH DESIGN

This research project combines design theories with practice-based research methods, positioning design as a tool for generating knowledge through experimentation, reflection, and iterative development. To explore Giant hogweed as an alternative cellulose resource, a combination of research methods was used to investigate the material from both a physical and systemic perspective.

MATERIAL-DRIVEN DESIGN

Material-Driven Design (MDD) is an approach where the material itself becomes the starting point for the design process, allowing its properties, behaviours, and potentials to define the outcome. In this project, MDD was used to investigate Giant hogweed as an alternative cellulose resource through creative material experimentation. Through an iterative process of exploring, analyzing, and developing the material, the research moves from a material proposal towards a potential design application.

REGENERATIVE DESIGN

Regenerative design is a design theory aiming to support the regeneration and resilience of ecosystems rather than simply reducing environmental impact. In this project, this framework guided the development of a speculative system that investigates how design can connect material production with ecosystem management, creating opportunities for more regenerative relationships between natural systems and the material economy.

DESK-BASED AND FIELD RESEARCH

Desk-based and field research were conducted to build contextual knowledge about invasive species management, material processing, and implementing potential resources.
Desk research informed the understanding of the issues and ecological impacts, existing practices and handling the toxic plants.
Field trips to SRV in Huddinge and individual visits to sites, based on the SLU Artdatabanken, in Stockholm provided insights into the impacts of invasive species, as well as current collection and management practices.

EXPERT INTERVIEWS

Interviews with experts complement the desk-based research and provide additional perspectives on invasive species management, material development, and systemic design approaches. The insights gathered helped validate and expand the initial research findings, while informing the further direction of the material proposal.

MATERIAL AND SYSTEM MAPPING

Material and system mapping were used as methods to understand Giant hogweed from both a material and contextual perspective.
Material mapping supported the investigation of the plant's components, processing possibilities, and potential material outcomes, helping to identify relevant directions for experimentation.
System mapping was used to visualize the broader flow of Giant hogweed biomass, including its removal, handling, disposal, and potential pathways for material utilisation. Together, these methods connected the experimental material research with the ecological, infrastructural, and social systems in which the material exists.

FROM WASTE TO RESOURCE

Currently, Giant hogweed is treated as an unwanted invasive species that requires costly and controlled removal, generating large quantities of biomass each year. As a result, its rapid growth is commonly perceived as a disadvantage, and its presence is framed as an ecological threat, an economic burden, and a long-term management challenge. Yet these same characteristics also reveal an overlooked material potential. Giant hogweed produces substantial amounts of cellulose-rich biomass within a single growing season, regenerating annually and accumulating large amounts of biomass.

To contextualize the potential, the productivity of Giant hogweed is compared with established cellulose crops, including industrial hemp and European forestry. Scots pine, one of Europe's most widespread tree species, typically produces approximately 3–5 tons of dry biomass per hectare annually, with a maximum of 7 tons, depending on site conditions.[19] Industrial hemp yields around 11–17 tons of dry biomass per hectare annually and is recognized for its high cellulose content.[20] In comparison, dense stands of Giant hogweed are estimated to produce approximately 5.2–15 tons of dry biomass per hectare per year.[21]
Although these values vary with environmental conditions, they demonstrate that Giant hogweed can generate biomass at a rate comparable to hemp and potentially up to twice that of productive forestry systems.

These productivity figures are estimations based on different sites and environmental conditions, and should be understood as indicative values for comparing biomass potential rather than exact production rates. The comparison is intended to support broader conclusions of biomass potential, as Giant hogweed currently occurs in dispersed and unmanaged populations. Furthermore, dence populations are associated with soil disruption and a decline in local species diversity. For this reason, this project focusses on existing waste to promote plant diversity and ecosystem stability on a site, rather than promoting single-species productivity.

[19] Lars Rytter, Morten Ingerslev, Antti Kilpeläinen, Piritta Torssonen, Dagnija Lazdina, Magnus Löf, Palle Madsen, Peeter Muiste, and Lars-Göran Stener, "Increased Forest Biomass Production in the Nordic and Baltic Countries – A Review on Current and Future Opportunities," Silva Fennica 50, no. 5 (2016): article 1660, https://doi.org/10.14214/sf.1660.

[20] John Finnan and David Styles, "Hemp: A More Sustainable Annual Energy Crop for Climate and Energy Policy," Energy Policy 58 (2013): 152–162, https://doi.org/10.1016/j.enpol.2013.02.046.

[21] Jan Pergl, Information on Measures and Related Costs in Relation to Species Included on the Union List: Heracleum mantegazzianum, Heracleum sosnowskyi, Heracleum persicum (Technical note prepared by IUCN for the European Commission, 2019), 20–21, https://invasivespeciesni.co.uk/wp-content/uploads/2021/10/IUCN-Datasheet-Heracleum-mantegazzianum-H.-sosnowskyi-H.-persicum-Hogweeds.pdf.

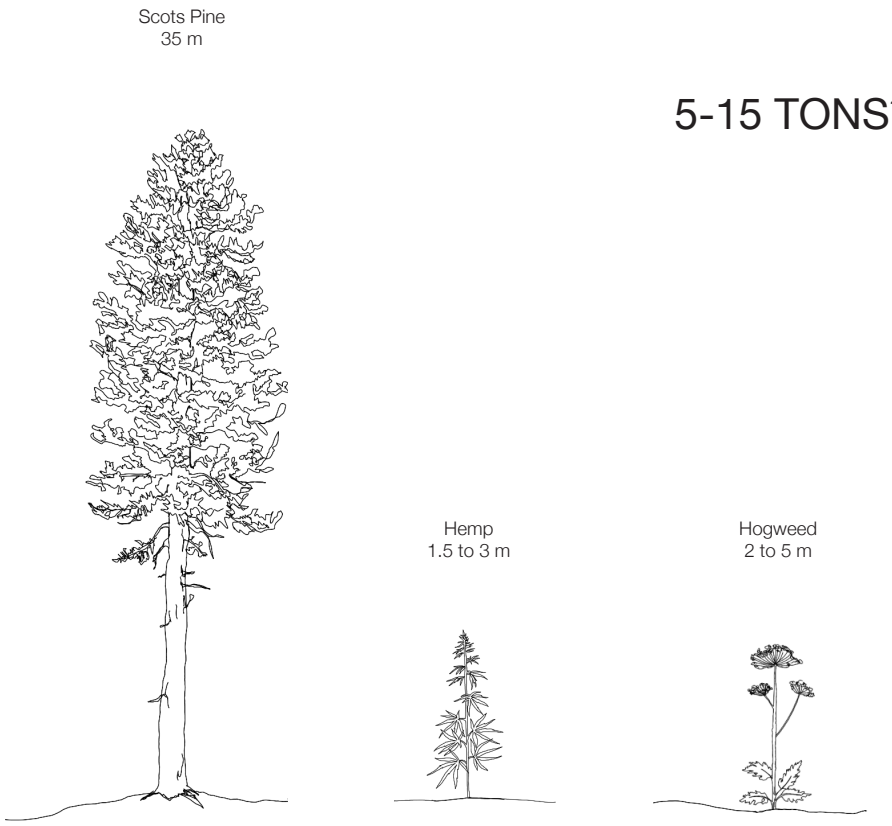


FIG. 9
Estimated annual biomass productivity per 1 ha
Source: Author

Building on the productivity comparison of Giant hogweed with conventional cellulose resources, this project expands the investigation from the material itself towards the broader context of how the species can be transformed from ecological waste to an alternative cellulose resource. The aim is to connect invasive species management with the cellulose industry creating the potential for both ecological and economic value *(Graphic 5)*.

From an ecological perspective, this approach connects the existing waste streams with the industrial production of cellulose, reducing the demand for virgin resources associated with deforestation, habitat degradation, and biodiversity loss. By using biomass that is already removed as part of ecosystem management, the new connection avoids additional land-use pressure while supporting the restoration and maintenance of ecosystems affected by invasive species.

From an economic perspective, this project aims to illustrate how invasive species biomass, currently considered a cost and waste stream, could gain new value within the existing material economy. Connecting invasive species management with the cellulose industry enables new value chains that create economic opportunities and potentially support ongoing management efforts.

Throughout the course of the degree project, the system expanded beyond Giant hogweed to explore broader material economies by connecting complementary biological resources. Rather than addressing Giant hogweed as an isolated ecological problem, the system incorporates sheep as a biological control measure. This creates a symbiotic co-production of two underutilized biological resources: cellulose from managed Giant hogweed biomass and wool from grazing animals.

Informed by the principles of regenerative design, this system demonstrates how ecological management can be transformed into regional resource production and supporting environmental protection and regeneration. *(FIG 10)*

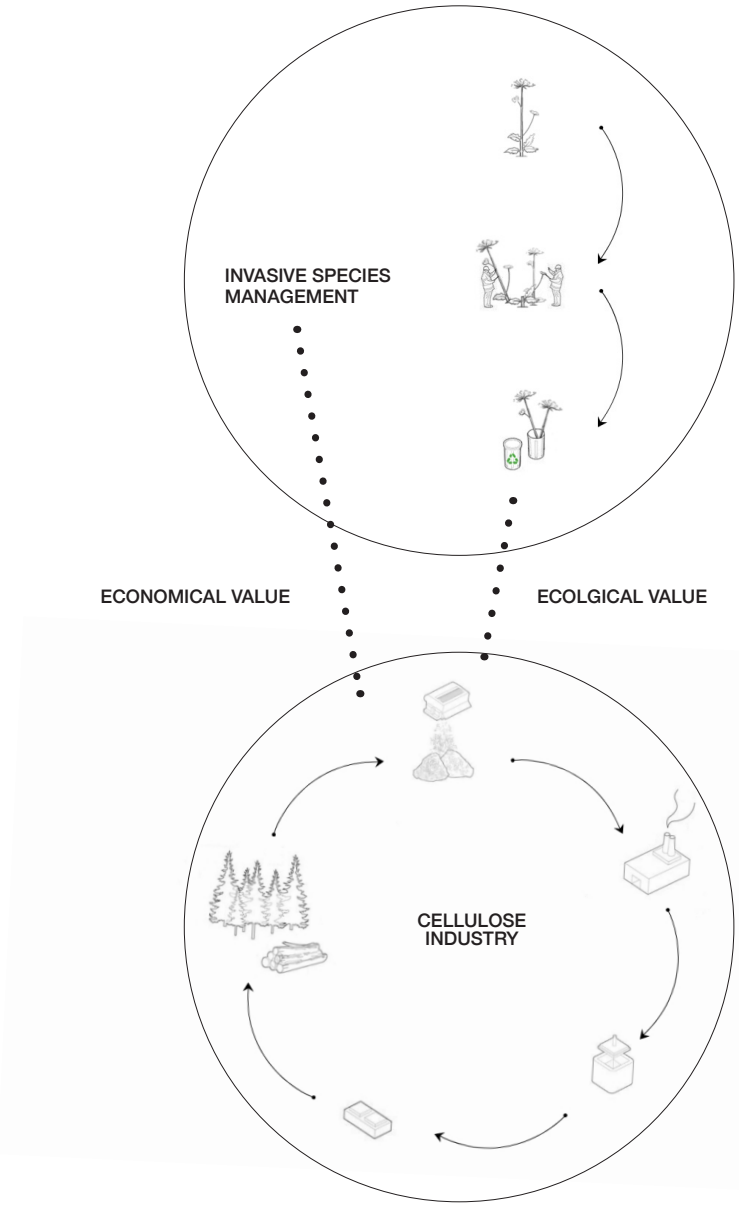


FIG. 10
Current system and intentions of integrating waste and creating ecological and economical value
Source: Author

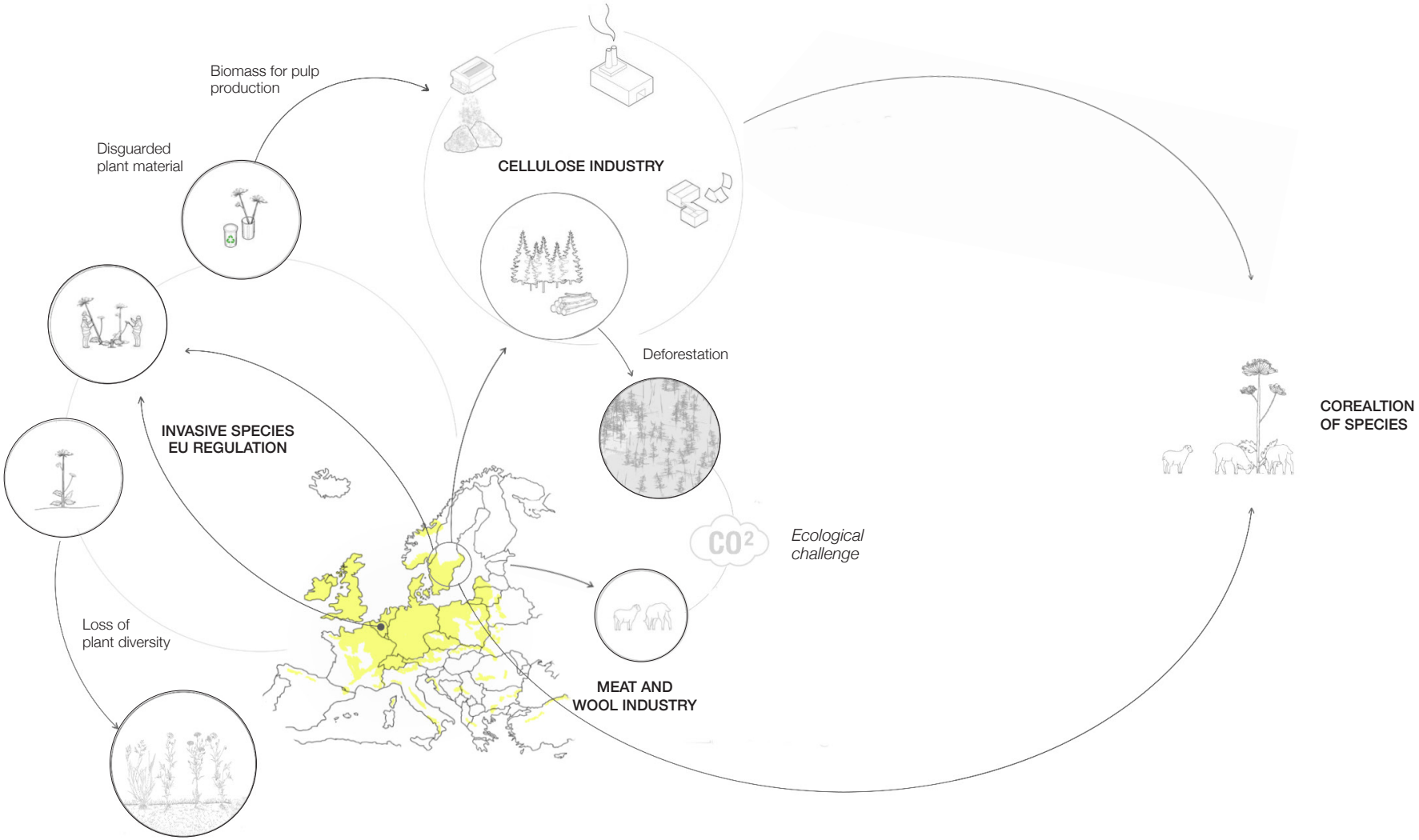


FIG. 11
Current system and intentions of integrating waste and creating ecological and economical value
Source: Author



FIG. 12
Fresh stem (left), retted stem (right) and residues
Source: Author



FIG. 13
Material mapping explorations
Source: Author

FROM RESOURCE TO MATERIAL

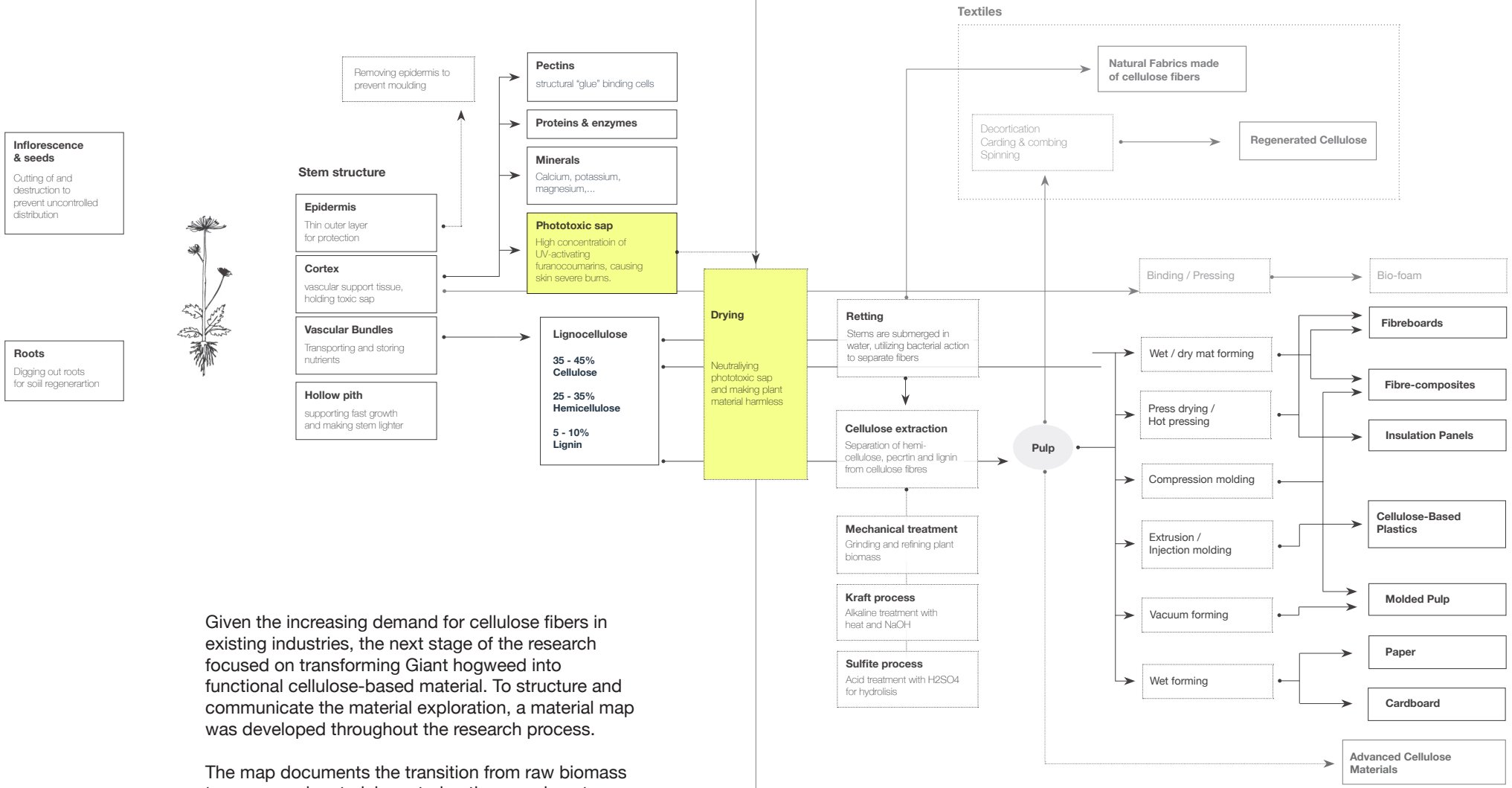
Starting with the practical design process, this project investigates the central question: *how to design with the toxic species Giant hogweed*.

The material exploration is informed by the principles of Material Driven Design (MDD), where the starting point is the material rather than a predefined product outcome. In the context of Giant hogweed, the approach aligns with Scenario 3 described in Material Driven Design: A Method to Design for Material Experiences by Elvin Karana (2015), which focuses on designing with a material proposal.[22] As Giant hogweed has not previously been explored as a processed material, the research begins by investigating its intrinsic properties, processing potential, and possible applications.

[22] Elvin Karana, Bahareh Barati, Valentina Rognoli, and Anouk Zeeuw van der Laan, "Material Driven Design (MDD): A Method to Design for Material Experiences," International Journal of Design 9, no. 2 (2015): 39, https://www.ijdesign.org/index.php/IJDesign/article/view/1965.

This design exploration begins with the analysis of the raw material. The stems of Giant hogweed consist of lignocellulosic biomass, containing cellulose fibers and lignin. They have a large hollow core surrounded by a soft, spongy pith and an outer layer of long fibers that provide mechanical strength and structural support. Similar to hemp, these fibers are located in the outer stem layer and contribute to the plant's rigidity.
The photo-toxic sap is distributed throughout the plant's vascular tissues. When the stem is cut the these tissues release the sap containing furanocoumarins, requiring careful handling during harvesting and processing.

The raw material used in this project was collected in autumn 2025 from locations around Stockholm. To prevent further spread of the invasive species, the flowering heads were removed and discarded in accordance with local management practices and the EU-Regulation before the stems were further processed.



Given the increasing demand for cellulose fibers in existing industries, the next stage of the research focused on transforming Giant hogweed into functional cellulose-based material. To structure and communicate the material exploration, a material map was developed throughout the research process.

The map documents the transition from raw biomass to processed material, capturing the experiments, treatments, manufacturing techniques, and resulting material possibilities. It serves a tool to map out the material potential of Giant hogweed cellulose and how different processing approaches influence its materiality and applications in different areas. Central to this project is the extraction of hogweed cellulose, as pulp provides a versatile foundation for developing a wide range of cellulose-based materials and applications, from molded fiber products to paper-like structures and composite materials. Therefore, this project focuses on developing cellulose pulp as the base material for material exploration and the design.

FIG. 14
Material Map from plant to material
Source: Author

The material exploration builds on established industrial pulping methods used in the paper industry to investigate how Giant hogweed can be integrated into existing cellulose production systems while adapting the processes to the characteristics of the raw material. After harvesting, the stems require drying under warm and well-ventilated conditions to reduce the activity of photo-toxic compounds and prevent mold formation before further processing. Once dried, the material can be ground and further processed into pulp.

The existing kraftprocess was used as a reference point. However, the composition of Giant hogweed enables the exploration of a lower-impact processing approach. Compared to wood, Giant hogweed contains less lignin, which makes the stem less dense and reducing the amount of chemical treatment required for fiber separation. Therefore, this research explores a soda-based pulping process as an alternative to the conventional white liquor commonly used in industrial paper production.

The pulping process results in two outputs: the cellulose pulp and a residual brown liquid. The resulting pulp exhibits strong fiber bonding, which is evident in the way the pulp forms clumps rather than separating into loose fibers. This indicates that the cellulose fibers remain long to interlock with one another, creating a strong fiber network. As a by-product, the process produces a brown solution, referred to in this project as brown liquor, named in relation to the black liquor generated in conventional paper production. Similar to black liquor, this solution contains dissolved lignin, hemicellulose residues, and residual chemicals from the pulping process. However, due to the lower chemical intensity of the developed process and the composition of Giant hogweed biomass, the brown liquor represents a less complex disposal.



Dried stems



Raw material



Pulping process



Refined material



Hogweed pulp



Brown liquor

FIG. 15
Pulping process of hogweed stems
Source: Author

FROM MATERIAL TO DESIGN

The material characteristics of the wet pulp informed the design direction by demonstrating the potential of hogweed pulp as a material to form self-supporting structures. Based on these findings, the material development explored two approaches for shaping the pulp: paper formation and three-dimensional molding. The first exploration focused on paper making to evaluate the stability, bonding capacity, and behavior of the raw fibers when formed into a flat cellulose structure. While the fibers were able to create a cohesive paper-like material, the end result was brittle and indicated a need to further explore forming methods that could enhance fiber bonding and structural integrity.

Inspired by paper molding techniques, the second exploration investigated compression molding as a method for creating dense, three-dimensional pulp structures. A two-part mold, consisting of a positive and negative form, was used to compress and dry the wet pulp mixture with a biodegradable binder. During drying, the compression process enhanced fiber bonding, resulting in durable and repeatable forms with structural integrity. Similar to conventional molded wood pulp manufacturing, this approach enables the production of consistent geometries through a repeatable molding process, while allowing control over key parameters such as material thickness, density, and shape.

The resulting material samples exhibit subtle color variations, ranging from light beige to darker beige and yellowish tones, influenced by the natural color of the stems. These variations reflect the raw origin of the material and contribute to a natural aesthetic.

From a structural perspective, hogweed cellulose and the resulting molded pulp are biodegradable and lightweight. Compared to conventional wood-pulp, the material contains longer cellulose fibers, resulting in improved fiber bonding and increased tensile strength. Beyond the mechanical properties, cellulose also exhibits functional characteristics that influence its potential applications. As a hydrophilic material, cellulose can absorb and regulate moisture, although prolonged exposure to high humidity may lead to material degradation. Due to the porous structure cellulose materials provide thermal regulation and sound absorption, making them suitable for interior applications and insulation. Additionally, the fibrous network has the ability to bind airborne particles such as dust, pollen, and micro plastics, creating potential for applications to improve air quality.



FIG. 16
Material samples:
hogweed pulp and paper
Source: Author



FIG. 17
Material samples:
Molded hogweed pulp
Source: Author



FIG. 18
Symbiotic materials: hogweed pulp and raw wool
Source: Author

THE ROLE OF WOOL

In this project, sheep are introduced as an component of the proposed material system, complementing hogweed by connecting ecological management with material production. Sheep contribute to the management of Giant hogweed through grazing, while simultaneously providing wool as a secondary resource.

Similar to Giant hogweed cellulose, wool is often overlooked within the existing material economy. In Sweden, wool is widely considered a by-product of the meat and dairy industries. Many sheep breeds raised for food produce coarse, short fibers that are unsuitable for conventional textiles. As a result, over 50% of Swedish wool is discarded or burned as waste, due to limited demand and processing costs, despite its considerable environmental footprint.[23]

Wool is a highly functional animal fiber known for its elasticity, resilience, and ability to absorb moisture and odors. Its fibrous structure traps air, which provides natural insulation, supports thermal regulation, and contributes to sound absorption. [24] Additionally, wool has the ability to bind airborne particles and certain volatile organic compounds. [25] These properties complement the qualities of hogweed pulp and offers possibilities to combine the lightweight and breathable materials in the design proposal.

[23] Axfoundation, "New Study Reveals Sustainability Benefits of Swedish Wool," June 15, 2023, https://www.axfoundation.se/en/news/new-study-reveals-sustainability-benefits-of-swedish-wool.

[24] Huitong Zhou, Lingrong Bai, Shaobin Li, Jiqing Wang, and Jon G. H. Hickford, "Wool: From Properties and Structure to Genetic Insights and Sheep Improvement Strategies," Animals 15, no. 19 (2025): 2790, https://doi.org/10.3390/ani15192790.

[25] International Wool Textile Organisation (IWTO), "Wool and Indoor Air Quality," accessed July 30, 2026, https://iwto.org/wellness/wool-indoors/.

DESIGN PROPOSAL

MATERIALS IN CONTEXT

[26] WWF, Living Planet Report 2024,7.

[27] American Society of Interior Designers (ASID), "Impacts of Extreme Weather on Interior Design Examined in New Research," March 28, 2024, https://www.asid.org/news/impacts-of-extreme-weather-on-interior-design-examined-in-new-research.

Climate change and pollution are two major drivers of biodiversity loss [26] and increasingly significant challenges for architecture, particularly in urban areas. Air pollution directly affects human health, as chemicals, VOCs, micro plastics, and other airborne particles accumulate in the environment and buildings. At the same time, shifting climate conditions are leading to warmer summer periods, rising indoor temperatures, and changing humidity levels.[27] These developments create new challenges for interior architecture. This project focuses on urban semi-public spaces such as cafés, restaurants, retail stores, and offices, where high occupancy levels, limited ventilation, and surrounding environmental conditions can intensify exposure to humidity, odors, VOCs, airborne particles, and noise, affecting both comfort and well-being.

These environments form the context of use for the design proposal, exploring how overlooked biological resources can actively respond to the emerging environmental challenges within interior spaces. The inherent material properties of hogweed pulp and wool make them particularly suited to these settings, where moisture regulation, sound absorption, and the capture of airborne particles and VOCs can contribute to the quality of those indoor environments.

As environmental conditions are distributed differently throughout a room, the placement of the material becomes an important design consideration. Warm air, moisture, and certain VOCs tend to accumulate near the ceiling, while airborne particles circulate throughout the interior. The following design proposal therefore explores the spatial integration of the materials to respond to varying environmental conditions and interior spaces.

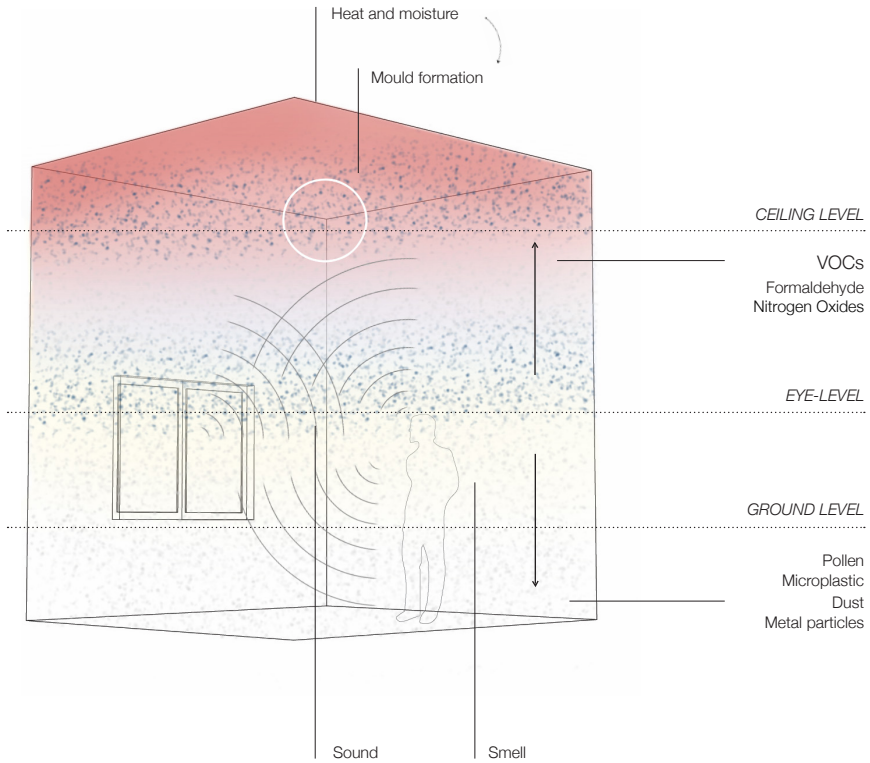


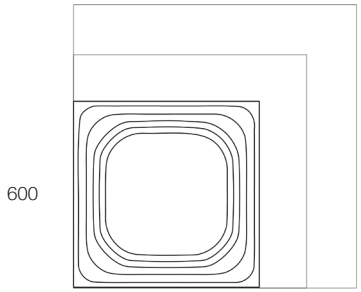
FIG. 19
Vertical zones and air pollutants in urban spaces
Source: Author

DESIGN PRINCIPLES

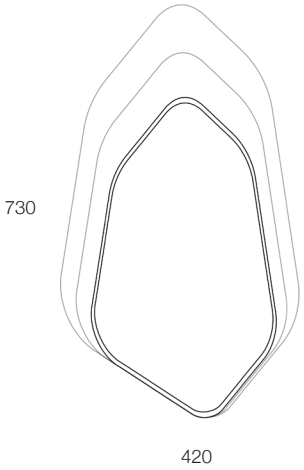
Pretty Toxic transforms Giant hogweed into a contemporary tiling system. The proposal consists of lightweight panels developed in two distinct geometries that adapt to a range of architectural contexts.

The design is informed by the material production process, where wet-forming the fibers through compression molding enables the repeated creation of consistent three-dimensional forms. Therefore, the shapes of the panels are based on the principle of monohedral geometry, in which a single repeating shape forms seamless patterns. Through repetition, the panels generate continuous surfaces that can expand across different scales and spaces, while simplify manufacturing, installation, and replacement.

The collection translates this principle into two complementary geometries. The square, measuring 600 × 600 mm, is compatible with existing architectural systems, designed to integrate into standard railing systems and suspended ceilings. In contrast, the leaf, measuring 730 × 420 mm, introduces a more complex geometry inspired by the leaves of Giant hogweed and is designed for suspended installations or wall-mounted compositions. When repeated, both geometries create a variety of seamless monohedral patterns that enhance their spatial presence while reflecting the spreading behavior of invasive alien species.



THE SQUARE



THE LEAF

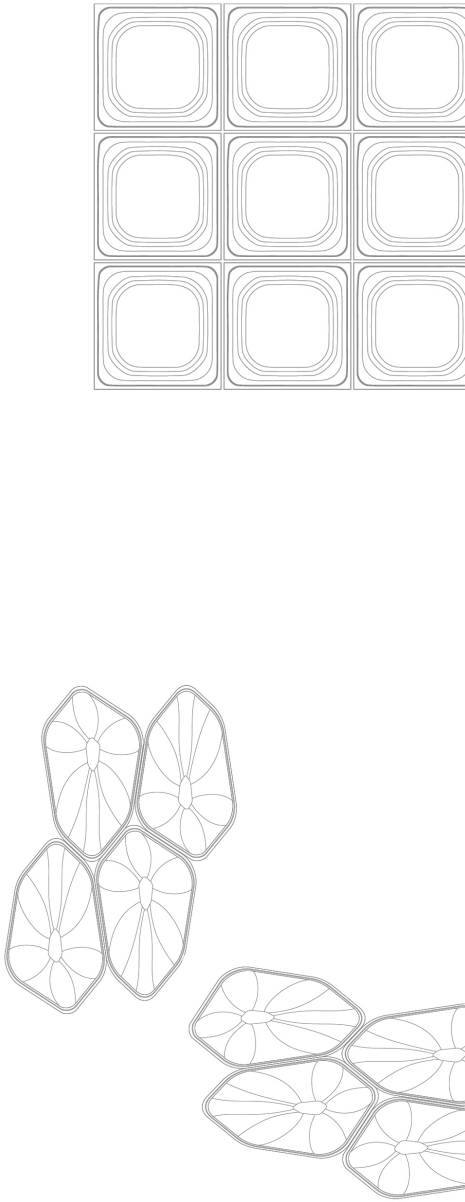
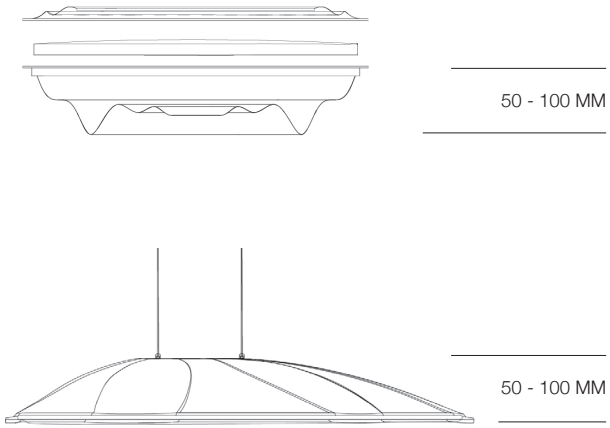


FIG. 20
Monohedral geometries and patterns
Source: Author

Each panel functions as a standalone acoustic element, based on three layers: a molded hogweed pulp backdrop, a low-density wool layer, and a structural frame also molded from pulp. The molded backdrop creates variations in cavity depth of 50–100 mm between the front and back layers, increasing the path of sound waves and improving acoustic performance. Due to the porous structure of the cellulose fibers, the pulp layer also supports moisture regulation and the capture of airborne dust and particles within the structure.

The low-density wool layer enhances sound absorption while supporting moisture regulation and particle binding within the structure. The structural frame, made from pulp, provides stability and encloses the wool core while maintaining the lightweight character of the panel.



Graphic 10
Cross section of the tiles
Source: Author

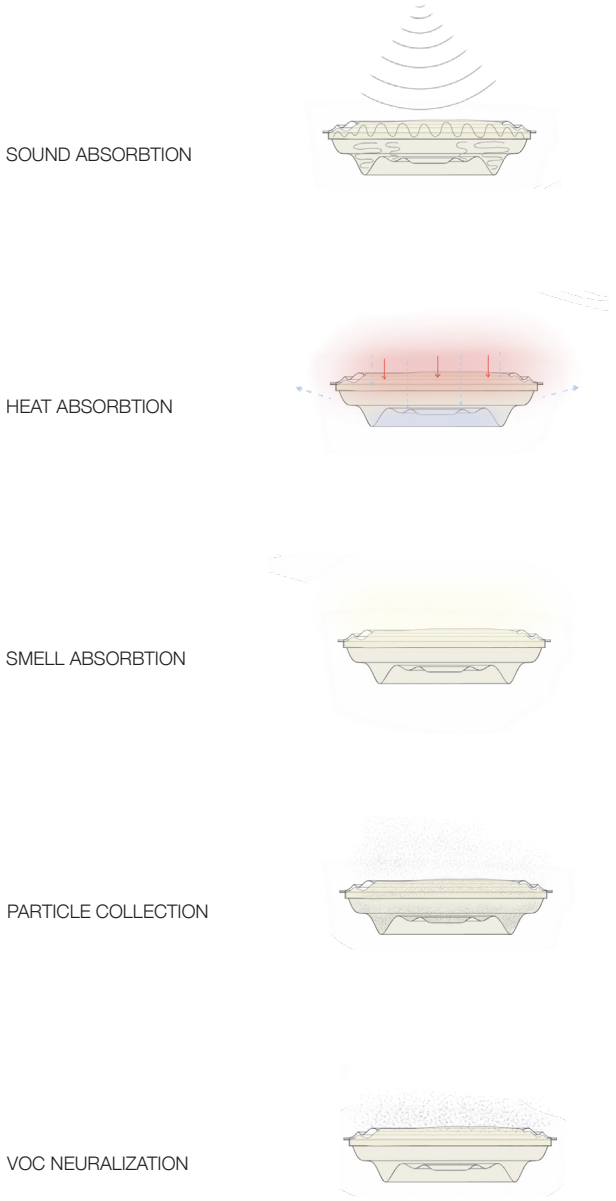


FIG. 21
Functions of the monohedral tiles
Source: Author

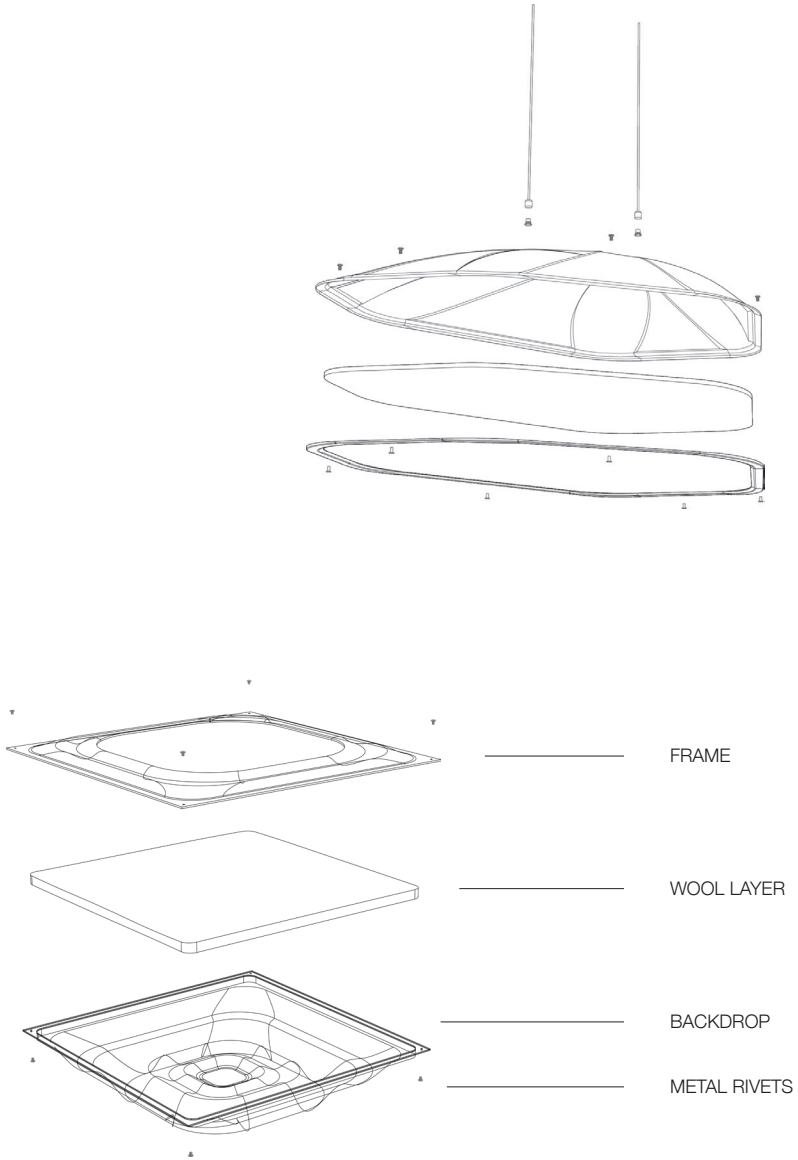


FIG. 22
Structure and layers of the tiles
Source: Author

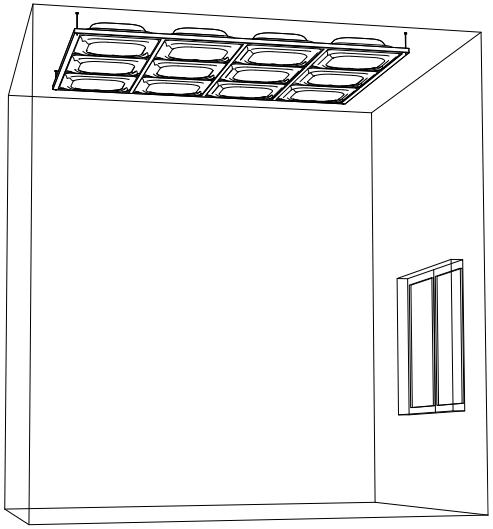
The layered construction follows a principle of design for disassembly, where the components can be separated, maintained, and replaced over time. The layers are assembled using long-lasting metal rivets that can be removed, reused, and recycled, allowing the individual material components to be recovered. The wool layer can be washed to restore its function, while the pulp components can be cleaned and maintained to prolong their performance.

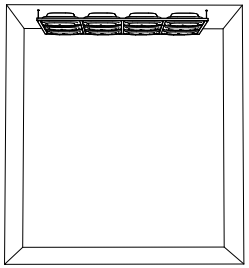
At the end of the product's life cycle, the individual materials can be reused or regenerated for future applications.





THE COLLECTION





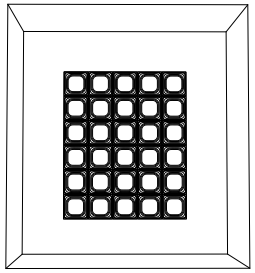
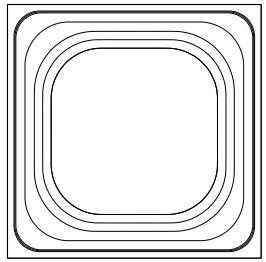


FIG. 23
Suspended ceiling grid and optional placements
Source: Author

THE GRID

The square panels are designed to integrate into existing standardized railing systems and suspended ceiling structures, allowing them to be applied in spaces with modular ceiling grids, such as offices, schools, and industrial environments. Alternatively, the panels can also be mounted directly onto ceilings or walls, providing flexibility across different architectural contexts and spatial conditions.
Positioned within the railing system at ceiling level, the panels provide a more passive and invisible integration into the interior. At this position, they respond to rising moisture and VOCs that accumulate near the ceiling, while also absorbing sound and smell.
As a wall-mounted application, the panels can function as continuous wall tiles or as standalone elements, focusing on absorbing sound, smell and particles that circulate at eye-level.



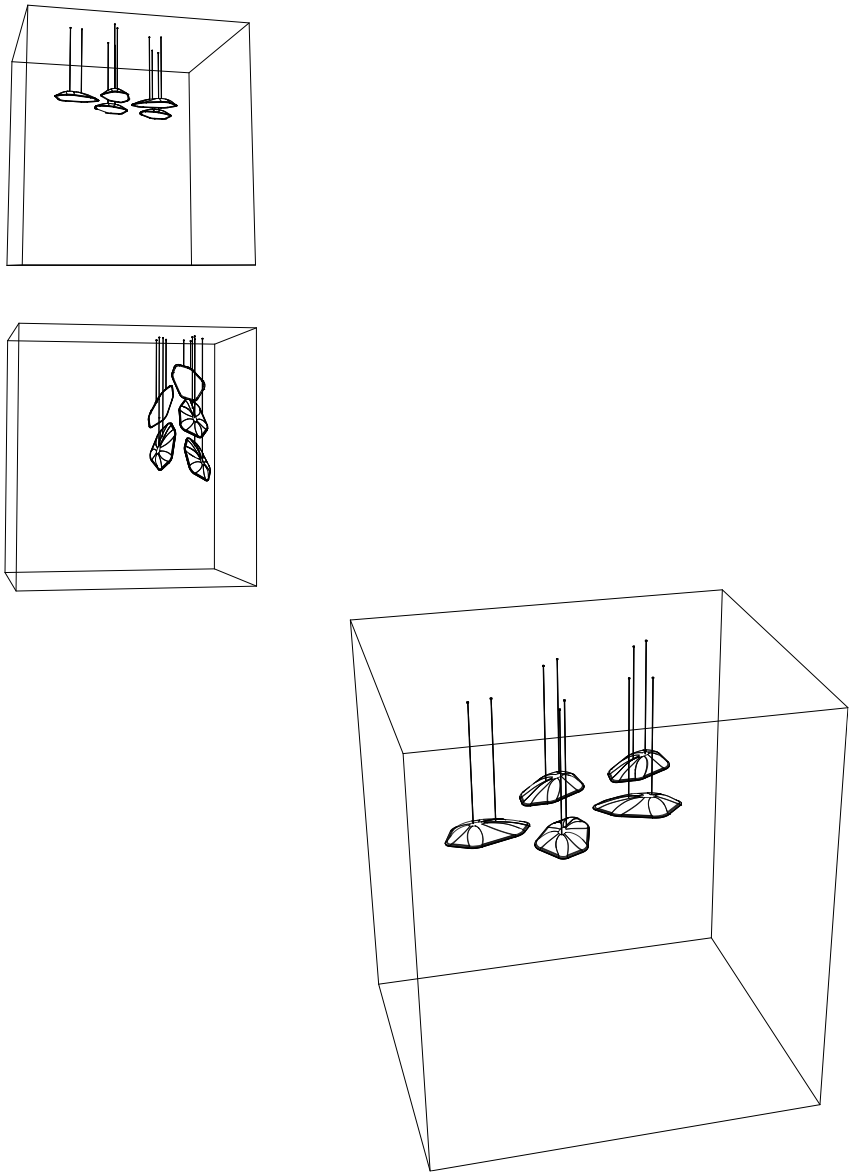
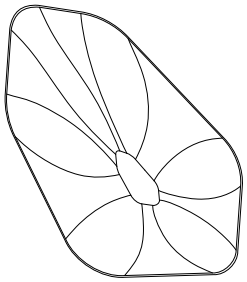


FIG. 24
Pending installation and arrangements
Source: Author

THE PENDANT

The leaf-shaped panels offer a third placement within the space, as the pendant installation introduces a more visible and decorative arrangement. In contrast to the more integrated square panels, the pendant tiles are designed as a spatial feature for contexts such as restaurants, cafés, and retail environments, where material expression and visual presence are central. The modular installation can be arranged in the center of a room or positioned within corners to adapt to different interior layouts. Through the string attachment system, the hanging height of the panels can be adjusted, allowing the installation to respond to different spatial conditions and user needs. By suspending the panels within the space, this application targets airborne particles, odors, and VOCs that accumulate at human height, while creating a decorative focal point within the interior.



REFLECTION

DESIGN FOR BIODIVERSITY

The underlining question of this design project was: how can design respond to biodiversity loss caused by invasive species, resource overexploitation, and pollution. Using the example of Giant hogweed, this project Pretty Toxic investigates how invasive species, commonly perceived as ecological threats, can be reconsidered as a potential material resource.
However, working with a toxic invasive species also revealed the complexity and contradictions involved in designing within ecological systems. Consequently, this project revealed that designing for biodiversity is not about extracting new resources from nature, but about understanding relationships between human activity, ecosystems, and production. Through a systemic approach, Pretty Toxic proposes that design can contribute to biodiversity by transforming managed biomass into functional material applications, reducing pressure on virgin resources, and supporting ecological management rather than creating new forms of resource extraction.

INVASIVE SPECIES AS RESOURCE

Throughout the development of the broader system that explores how Giant hogweed can become a resource and material stream, the question emerged whether the plant could become a resilient resource within a future agricultural system. But, the same characteristics that make Giant hogweed a potential biomass resource are also the reasons for its ecological impact and ability to disrupt existing ecosystems. Giant hogweed forms dense and persistent stands, which rapidly displaces native vegetation, reduces plant diversity, and alter habitats that support local species even in a controlled setting.

Therefore, the project does not propose the cultivation of Giant hogweed as a new agricultural resource. By introducing an invasive species into controlled production systems could create new ecological risks. Large-scale cultivation of a single invasive species would risk reproducing the same challenges associated with the current industrial agriculture and reinforce the same extractive logic that contributes to ecological damages and biodiversity loss.

Instead, Pretty Toxic explores the potentials of existing materials, that are overlooked. The project aims to contribute to a shift towards recognizing and utilizing existing material streams rather than creating new systems of extraction. This is why, it proposes a system where material production can support and impact, rather than conflict with, biodiversity management.

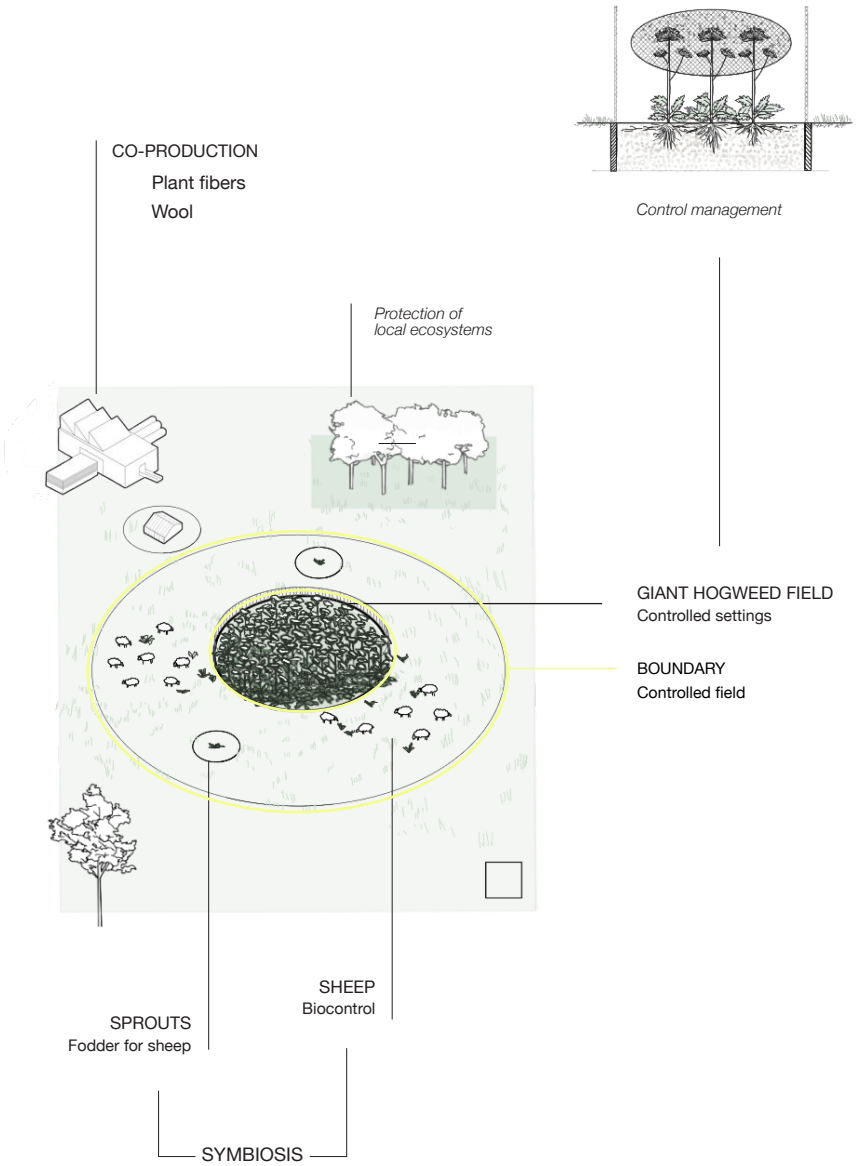


FIG. 25
Cultivation scenario of Giant hogweed
Source: Author

MATERIAL AND DESIGN

Alternative cellulose resources offer opportunities to reduce dependency on conventional materials and expand the diversity of cellulose sources available for design and industry. Within this context, this project explored the development of hogweed pulp to transform an overlooked waste stream into a material with functional and aesthetic qualities. The material samples revealed the potentials of hogweed fibers to become a usable and versatile material to produce.

Based on the approach of Material Driven Design (MDD), the developed material is positioned within Scenario 3 and therefore a first material proposal rather than a final solution. In this project, the research and design process focused on understanding the raw material's inherent properties and possible applications through iterative experimentation. The resulting design proposal represents the exploration of how the material could be translated into an application and it`s aesthetic and functional qualities.

The current material development represents a intermediate outcome that requires further development to improve material consistency, durability, moisture resistance, scalability, and compatibility with industrial processing methods. Further testing of the material's inherent properties would support the development of specific material functions. Similarly, the tile design represents an initial exploration of how this material could be applied within interior environments and requires further developing and testing regarding performance, manufacturing processes, and architectural integration.

FUTURE POSSIBILITIES AND LIMITATIONS

This project demonstrates the potentials of Giant hogweed as an alternative cellulose resource, but presently certain limitations exist in working with an invasive species. A fundamental challenge is the absence of an established material flow. For this project, the Giant hogweed biomass was sourced and collected personally through controlled harvesting. The material preparation and processing carried out specifically for experimental purposes. However, for Giant hogweed to become part of a future material system, structures for harvesting, transportation, processing, and collaboration between ecological management and material industries needs to be developed. This would require integrating invasive species management into a coordinated system rather than distributed between multiple actors such as municipalities, private landowners, and other local stakeholders.

Furthermore, the use of Giant hogweed as a material resource requires careful consideration of the current EU-Regulation, toxicity, and ecological responsibilities. Due to its photo-toxic properties, safe handling procedures throughout harvesting, processing, and material production are essential. Guidelines regarding the handling, transport, processing, and potential use of invasive biomass are necessary to ensure that material utilization does not contribute to further spread or ecological damage. Scaling up this project would therefore depend on collaboration between researchers, municipalities, environmental authorities, and industries to establish responsible frameworks for managing and utilizing this resource.

Despite these limitations, this project demonstrates the possibilities of design and material research to identify alternative resources and create local material cultures. By working with a species that is often perceived as a threat, Pretty Toxic contributes to a broader discussion on how we define resources and waste, and how overlooked materials can be used.
Ultimately, Pretty Toxic repositions design as a tool to support ecological management and biodiversity by creating new relationships between environmental challenges, material resources, and human needs.

CONCLUSION

As biodiversity is declining, Pretty Toxic connects design with environmental protection by reframing the invasive alien species Giant hogweed as a potential material resource. Combining Material Driven Design and Regenerative Design, the project explores how a toxic invasive species can be transformed into functional materials and applications that create new relationships between biodiversity, material production, and human needs.

The project results in three interconnected outcomes: a system proposal, developed material, and a design application.

The system proposal connects ongoing invasive species management with existing material industries by linking ecological management practices with the cellulose and wool industries. It demonstrates how biomass that already requires removal could gain new value as a resource with ecological and economic potential.

The material development investigates the processing of Giant hogweed stems into pulp and a lightweight, moldable cellulose-based material. Through wet-forming experiments, the project explored the manufacturing process to form repeatable structures that could be further adapted within existing production methods. The molded material outcome represents an initial proposal that requires further

The design proposal translates the material research into a modular tile system, combining hogweed pulp and wool to create adaptable and functional interior elements with potential contributions to indoor environments. The materials can support spatial conditions through functions such as sound absorption, temperature regulation, and the ability to interact with indoor air by capturing particles, reducing odours, and contributing to the neutralisation of VOCs. The prototypes function as a first design proposal rather than a finalized product. Their purpose is to communicate the potential of the material while demonstrating the areas that require further investigation, including material performance, processing methods, and material sourcing.

Pretty Toxic contributes to design research by questioning how we define resources and waste and proposing alternative relationships between biodiversity, materials, and production. Ultimately, The project highlights how design can support ecological management by creating new narratives around overlooked resources and exploring future material cultures.

USE OF AI

In this thesis, AI tools were used throughout the project as a supportive resource for the search of specific research input, structural editing and language refinements of the written manuscript, such as altering formulations and refining technical terms. AI was not used to replace critical thinking or creative development, but as a tool to support the organization and refinement of ideas. All theoretical and practical decisions, material experiments, design concepts, and final outcomes were developed and evaluated through my own design practice and critical judgment.

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APPENDIX



FIG. 26
Exhibition of Pretty Toxic at
Vårutställning 2026, Konstfack University
Source: Author

THE EXHIBITION

Pretty Toxic was exhibited at the annual Konstfack Vårutställning 2026 from 13–26 May. The exhibition presented all graduation projects under the theme Connecting the Dots and attracted approximately 13,000 visitors during the exhibition period.

The exhibition was arranged to guide visitors through the project, presenting a clear progression from raw plant matter and material development to the final design proposals. The material process was displayed chronologically on a black podium, illustrating the transformation of Giant hogweed into a cellulose-based material, with labels guiding visitors effectively through the project. The dried Giant hogweed plant served as a central focal point, attracting visitors and introducing the project's topic. Suspended above the podium, the pendant installation acted as a strong visual landmark, highlighting the material qualities and spatial potential of the final design proposal.

During the exhibition, the project received very positive feedback from visitors, particularly for its material aesthetics and the proposal to transform Giant hogweed into a valuable design resource.
As the plant is widely recognized as an invasive species in Sweden, many visitors immediately connected with the topic. Despite its known toxicity, visitors readily interacted with the processed material samples, demonstrating a shift in perception.

The most common questions concerned the project's future development, material scalability, and the sourcing and processing of Giant hogweed biomass, reflecting strong interest in future developments of this project.





FIG. 27
Exhibition of Pretty Toxic at
Vårutställning 2026, Konstfack University
Source: Author



FIG. 28
Close up of the square tiles
Source: Author



FIG. 29
Close up of the pendant
Source: Author



Master thesis

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